Threat Hunting Practice

1. Create Hypothesis:-

An unauthorized administrative access may have occurred in the system.
This activity could indicate misuse of privileged accounts or credential compromise.

2. Open Elastic Security (Kibana):-

- Steps to perform:-

  ->Open **Kibana.**

  ->Go to:
  Security → Discover.

  ->In search bar, type:-

  - event.code: 4625

3. Log Analysis:-

Now check these fields in Kibana:

- **user.name** → which account is targeted
- **source.ip** → attacker IP.
- **@timestamp** → time (very important)
- **logon.type** → type of login

4. Table:-

Timestamp	User	Event ID	Notes
02:10 AM	admin	4625	Failed Login
02:11 AM	admin	4625	Repeated attempt
02:12 AM	admin	4625	Possible Brute Force

5. Threat Intelligence Check:-

- Steps:-

1. Open **AlienVault OTX**
2. Search:
    - Suspicious IP from logs
3. Check:
    - Malicious activity
    - Reputation

The suspicious IP address was checked using AlienVault OTX.

The results indicated potential malicious activity associated with the IP, confirming the likelihood of unauthorized access.

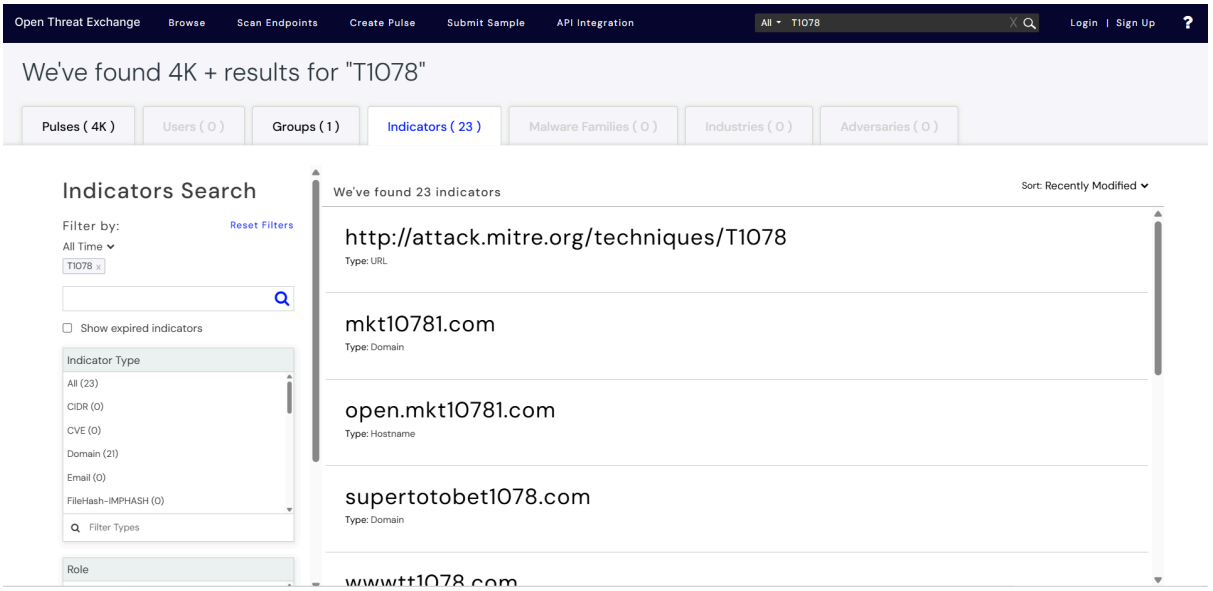
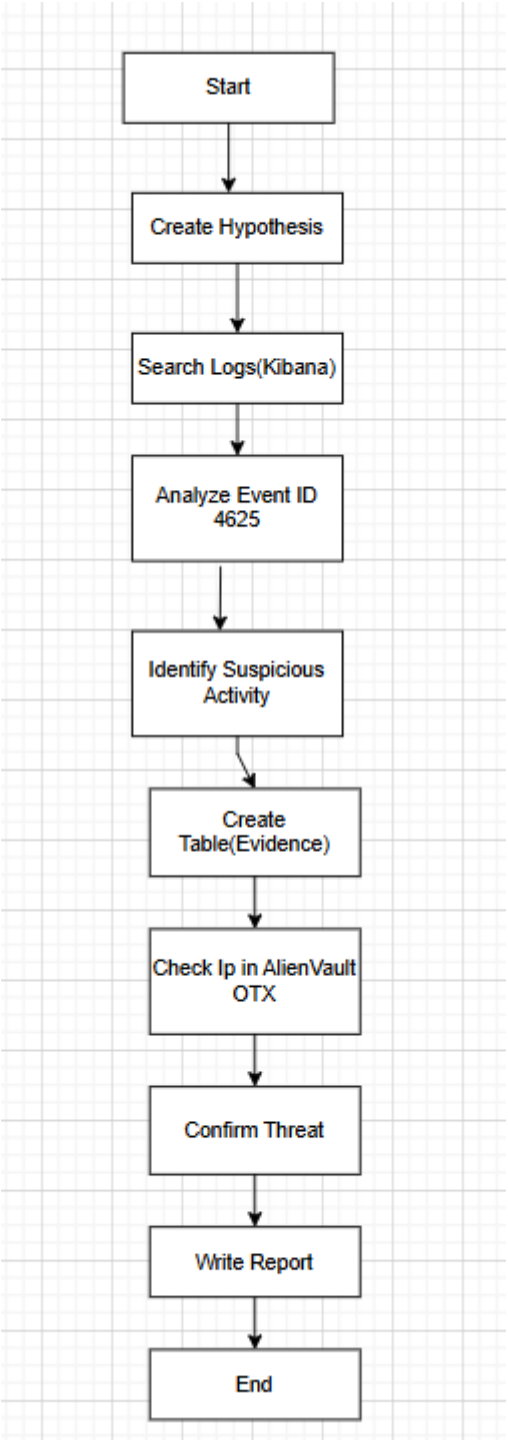


Figure 1: OTX search result page

6. Flowchart:-



7. Conclusion:-

A threat hunting investigation was conducted to detect unauthorized access attempts.
Event ID 4625 logs were analyzed in Kibana, revealing multiple failed login attempts targeting a user account.

The activity occurred repeatedly within a short time frame, indicating a potential brute-force attack.
The source IP address was further analyzed using AlienVault OTX, which showed suspicious indicators.
This behavior aligns with MITRE ATT&CK technique T1110 (Brute Force), where attackers attempt to gain access using multiple password attempts.
The findings suggest a possible attack on the system requiring immediate monitoring and response.